



CONTACT

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EDUCATION

- FRIEDRICH ALEXANDER UNIVERSITY**
Erlangen, Germany
 - Master of Science in Physics (Astrophysics and Astro-Particle Physics)
2022 - 2026
- MUMBAI UNIVERSITY**
Mumbai, India
 - Bachelor of Science in Physics
2018 - 2021

SKILLS

- Python / Jupyter / Linux / Git
- Scientific Computing / Data Analysis
- Scikit-learn / XGBoost / LightGBM
- GPyTorch / MLflow
- ML / Transformers
- VLA Models / World Models
- Isaac Sim / Omniverse / ROS 2
- Franka FR3 Duo / GELLO
- Robotic Manipulation
- Copilot Agents / AI Automation
- PyVISA / Instrument Automation
- Calibration / Systematics
- PCA / Anomaly Detection / QC
- Power BI / Superset / Looker
- Leadership / Feasibility Analysis

LANGUAGES

- English (C1)
- German (~B1)
- Marathi (Native)
- Hindi (Native)

SAUJANYA DUBALE

saujanyadubale.github.io | github.com/saujanyadubale | linkedin.com/in/saujanya-dubale

I am a constant learner fuelled by curiosity. I think in first principles: I observe, analyse, identify problems, debug, and conclude. I seek out meaningful problems and create solutions that deliver value and drive positive change.

WORK EXPERIENCE

Siemens Healthineers

Working Student | Embodied AI & Robotic Manipulation R&D

04/2026 - 10/2026
Forchheim, Germany

- Developing bimanual manipulation workflows for the Franka FR3 Duo, integrating NVIDIA Isaac Sim, GELLO teleoperation, ROS 2-oriented control preparation, and simulation-based robotic learning.
- Evaluating Vision-Language-Action (VLA) models, including OpenPI π 0.5, GR00T N1.7, and SmolVLA, for perception-guided manipulation, inference performance, embodiment compatibility, and task execution.
- Developing fine-tuning pipelines using robot-specific demonstrations, cross-embodiment transfer, and different adaptation strategies, including full-policy training, selective layer fine-tuning, and LoRA.
- Investigating complementary learning approaches, including Action Chunking Transformers (ACT), Diffusion Policy, supervised residual correction, and reinforcement learning (SAC), to improve bimanual action prediction and control.
- Proposed a predictive-verification framework combining VLA-generated actions with world models, stochastic rollouts, outcome critics, and receding-horizon control. Explored Cosmos-Predict and TD-MPC2-style approaches for uncertainty-aware planning and future-state prediction.

Siemens Healthineers

Working Student | AI Automation & Knowledge Management

03/2025 - 04/2026
Erlangen, Germany

- Built multiple Copilot-based agents for internal document search, question answering, and knowledge reuse across technical and organisational content.
- Developed an interview-driven support workflow combining multilingual English, German, and Chinese audio/video transcription with structured response generation for recurring customer and stakeholder questions.
- Curated and restructured large internal knowledge collections by removing duplicate, outdated, noisy, and low-value material, improving retrieval quality and the reliability of generated answers.
- Supported data validation, recurring reporting, and decision-oriented summaries by translating complex datasets and stakeholder inputs into clear, reusable information.
- Managed the editorial workflow for the CBL Connect newsletter and maintained an interactive Task-Based Services intranet hub, improving access to team updates, services, and technical resources.

Erlangen Centre Of Astroparticle Physics (ECAP) (FAU)

Hilfs-Wissenschaftler and Student Researcher (Thesis)
Grade: 1.0

06/ 2024 - Present
Erlangen, Germany

- Built an end-to-end optical detector-characterization pipeline to assess recovered ANTARES PMTs for possible reuse in SWGO.
- Designed and validated a light-tight measurement system combining a xenon source, monochromator, calibrated photodiode, picoammeter, and motorized rotation stage.
- Automated wavelength and angular scans in Python, integrating instrument control, parameter readback, dark-current acquisition, metadata logging, and reproducible outputs.
- Diagnosed lamp drift, dark-current relaxation, low-current sensitivity, and the 400 nm monochromator transition, distinguishing instrumental artifacts from genuine detector behaviour.
- Analysed 124 PMT spectra using robust statistics, spectral metrics, PCA, and Isolation Forest; retained 114 measurements for interpretation and flagged 10 for targeted diagnosis or remeasurement.

WORK EXPERIENCE

Promeos GmbH

Working Student | AI R&D (SMEs) and Office Associate

09/2023 - 03/2024
Nürnberg, Germany

- Assisted the CEO with research on practical AI implementation opportunities for small and medium-sized enterprises.
- Prepared cost analyses for AI adoption, comparing potential tools, implementation effort, expected value, and operational feasibility.
- Designed marketing and business communication materials, including brochures, logos, and visual graphics for company presentations and outreach.

Board of European Students of Technology

Grants Department | Data and Business Analysis Department

01/2025 - 02/2026
Europe

- Grants Co-ordinator for Advocacy Team Project (ATP) : Primary contact point between Grants Department and ATP team
- To study and write the ATP grant application that BEST submitted to European Youth Foundation
- Responsible for keeping track of what we promised in the application and communicating the needs to the Local BEST Group and the team.
- Designed and delivered sessions and supported Europe-wide dissemination of program results through in-person delivery and partner outreach
- Co-created an advocacy toolkit (templates, guidance, resources) during a multi-day intensive, enabling local branches across Europe to integrate advocacy into standard operations.
- Restructuring and updating of internal knowledge management platform for Data and Business Analysis Department
- Creation of guidelines for uniform use of Google Workspace for documentation and Apache Superset and Google Looker for visualization

3foldtec

Administrative & Operations Assistant

11/2021 - 08/2022
Thane, India

- Managed production inventory, keeping an overview of production and assimilating/negotiating supplies.
- Managed top 10 clients relations with understanding their needs and finding new ones.
- Handling demands and requirements of clients based on market needs.

High Serve Society & Knowter - The Knowledge Barter

Founder

03/2020 - 08/2024
Mumbai, India

- High Serve Official was a sports management startup dedicated to delivering unparalleled experiences to hardworking and passionate players. Gave over 2000+ athletes a starting point to showcase their talent.
- Knowter was a skill/knowledge-sharing platform built around relatability, accessibility, and user freedom.
- Built an MVP matchmaking platform connecting people via freely available resources; onboarded 80+ early users.
- Formerly Scienspiration: an outreach initiative bridging underprivileged and privileged students through context-based science learning. Reached 1,000+ students via schools/colleges, delivering interactive sessions and science learning activities.

CERIFICATIONS

Application of AI in Healthcare

University of Lisbon

Sustainable Transport & Green Logistics

National Technical University of Athens

Next-Generation Materials & Advanced Manufacturing

University of Ljubljana

Silicon Thin Film Solar Cells

Ecole Polytechnique

Density Functional Theory

Ecole Polytechnique

ORGANIZATIONS

Board of European Students of Technology (Erlangen) e.V

Full Member

Full member of BEST Erlangen representing FAU across Europe; vote/propose governance changes, contribute to international teams (Grants, Advocacy, Data/Business Analysis), and design & deliver sessions.

Space Gene, FAU

Vice President

Lead teams and workflows, drive event ideation and execution, liaise with professors and sponsors, run meetings, and represent the organization externally.

Leo Club of Vashi Gold (NGO)

Member (Service Volunteer)

Planned and delivered community service initiatives end-to-end, leading multiple events and coordinating volunteers to drive measurable local impact.